Cross-Script Alignment Prompting for Offensive Language Detection

Prompt-driven Detection of Offensive Urdu Language: Cross-Script Robustness in LLMs [category 2]

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Motivation

The proliferation of tech-facilitated abuse on Pakistani social media platforms occurs predominantly in code-switched Roman Urdu. Standard NLP moderation tools often fail because they are trained on clean, standardized Urdu script and cannot handle the extreme spelling variations (e.g., *nahi*, *nai*, *nhi*) inherent to informal Roman Urdu. While recent advancements, such as the prompt-driven detection established by Maab et al. (2026), achieve high accuracy on standard Urdu script, they experience a significant performance degradation when applied to noisy chat text. Bridging this gap without requiring resource-intensive model fine-tuning is critical for developing scalable, robust moderation systems for low-resource linguistic environments.

Problem Statement

This project re-implements the baseline zero-shot and few-shot prompting strategies for offensive language detection as proposed by Maab et al. (2026). It then introduces a novel intervention: Cross-Script Alignment Prompting. The core research question is: *Does augmenting LLM prompts with a phonetic alignment dictionary (mapping Roman Urdu spelling variants to standard Urdu script) improve offensive language detection on noisy social media text compared to standard few-shot prompting?*

Methodology

- **Phase 1: Baseline Reproduction:** Replicate the original methodology using TF-IDF with a linear SVM as a classical lower bound, alongside zero-shot and standard few-shot prompting (providing 3-5 standard examples without script alignment) using an open-weight model like Llama-3-8B.

- **Phase 2: Novel Intervention:** Implement Cross-Script Alignment Prompting. The LLM is instructed to explicitly map Roman Urdu slang and abbreviations to their standard Urdu script equivalents (e.g., *kch* to کچھ) within the context window before executing the classification task.

- **Phase 3: Evaluation:** Evaluate models using Macro-F1 (primary metric), Precision, and Recall using scikit-learn across three distinct test domains: clean Urdu script, standard Roman Urdu, and a custom spelling-variation dataset. Computational latency will also be recorded.

- **Phase 4: Ablation Study:** Systematically vary the size of the injected alignment dictionary (e.g., 0, 5, and 15 mapping examples) to determine the optimal balance between classification accuracy and token inference cost.

- **Phase 5: Error Analysis:** Conduct a qualitative manual review of 50 misclassified comments from the optimal model to categorize failures into a formal taxonomy (e.g., missed sarcasm, English slang confusion, or transliteration hallucination).

Dataset Details

- **RUHSOLD (Roman Urdu Hate Speech and Offensive Language Dataset):** A dataset of approximately 10,012 tweets, utilized for testing standard Roman Urdu performance.

- **Akhter et al. (2020) Dataset:** Contains social media comments in both Urdu script and Roman Urdu; utilized to test the baseline's performance on clean, standardized text.

- **Custom Spelling-Variation Set:** A newly curated test set of 300 Roman Urdu comments, intentionally injected with high-frequency spelling variants and English mix-ins by the research team to stress-test cross-script model robustness.

References

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